Optimization and Curve Sketching

ANSWER KEY

Section 1: Critical Points and First Derivative Test

1 Find all critical points of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. $x = 1$ **and** $x = 2$ ✓

b. $x = -1$ and $x = -2$

c. $x = 3$ and $x = 0$

d. $x = 2$ only

$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2) = 0 \Rightarrow x = 1$ and $x = 2$.

2 Classify the local extrema of $f(x) = 2x^3 - 9x^2 + 12x - 5$.

a. **Local max at** $x = 1$**; local min at** $x = 2$ ✓

b. Local min at $x = 1$; local max at $x = 2$

c. Inflection point at $x = 1$

d. Local max at $x = 2$ only

$f'(x)$ is positive for $x < 1$, negative for $1 < x < 2$, and positive for $x > 2$. By the First Derivative Test, $x = 1$ is a local max and $x = 2$ is a local min.

3 Find the absolute maximum of $g(x) = 4x - x^2$ on the closed interval $[0, 3]$.

a. 4 ✓

b. 3

c. 0

d. 5

$g'(x) = 4 - 2x = 0 \Rightarrow x = 2$. Test critical point and endpoints: $g(0) = 0, g(2) = 4, g(3) = 3$. The absolute maximum is 4 at $x = 2$.

Section 2: Concavity and the Second Derivative Test

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4 Determine the interval(s) where $f(x) = x^4 - 4x^3 + 10$ is concave upward.

a. $(-\infty, 0) \cup (2, \infty)$ ✓

b. $(0, 2)$

c. $(2, \infty)$

d. $(-\infty, 2)$

$f'(x) = 4x^3 - 12x^2$. $f''(x) = 12x^2 - 24x = 12x(x-2) > 0 \Rightarrow x < 0$ or $x > 2$.

5 Find the x-coordinates of all inflection points of $y = x^3 - 6x^2 + 9x$.

a. $x = 2$ ✓

b. $x = 1$ and $x = 3$

c. $x = 0$

d. $x = 4$

$y' = 3x^2 - 12x + 9$. $y'' = 6x - 12 = 0 \Rightarrow x = 2$. Concavity changes from negative to positive at $x = 2$.

6 If $f'(3) = 0$ and $f''(3) = -5$, what does the Second Derivative Test conclude about $x = 3$?

a. f **has a local maximum at** $x = 3$ ✓

b. f has a local minimum at $x = 3$

c. $x = 3$ is an inflection point

d. The test is inconclusive

Since $f'(3) = 0$ and $f''(3) < 0$, the graph is concave down at the critical point, indicating a local maximum.

Section 3: Applied Optimization: Perimeter, Area, and Volume

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7 A farmer has 120 m of fencing to enclose a rectangular field adjacent to a river (no fence needed along the river). What dimensions maximize the enclosed area?

a. **Width** $= 30$ m**, Length** $= 60$ m ✓

b. Width $= 40$ m, Length $= 40$ m

c. Width $= 20$ m, Length $= 80$ m

d. Width $= 25$ m, Length $= 70$ m

Perimeter: $2w + l = 120 \Rightarrow l = 120 - 2w$. Area: $A(w) = w(120 - 2w) = 120w - 2w^2$.
$A'(w) = 120 - 4w = 0 \Rightarrow w = 30$ m, $l = 60$ m.

8 An open-top rectangular box is made from a 16 by 10 inch cardboard sheet by cutting equal squares of side x from each corner and folding up the sides. What x maximizes the box volume?

a. 2 inches ✓

b. 3 inches

c. 1.5 inches

d. 2.5 inches

$V(x) = x(16 - 2x)(10 - 2x) = 4x^3 - 52x^2 + 160x$.
$V'(x) = 12x^2 - 104x + 160 = 4(3x^2 - 26x + 40) = 4(3x - 20)(x - 2) = 0$. Since $x < 5$, $x = 2$ inches.

9 A closed cylindrical can must hold 1000 cm^3 of liquid. What relationship between height h and radius r minimizes total surface area?

a. $h = 2r$ ✓

b. $h = r$

c. $h = 4r$

d. $h = \pi r$

$S = 2\pi r^2 + 2\pi rh$. With $V = \pi r^2 h = 1000 \Rightarrow h = 1000/(\pi r^2)$. $S(r) = 2\pi r^2 + 2000/r$.
$S'(r) = 4\pi r - 2000/r^2 = 0 \Rightarrow 4\pi r^3 = 2000 = 2V = 2\pi r^2 h \Rightarrow h = 2r$.

Section 4: Economics and Distance Optimization

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10 A company's revenue and cost functions are $R(x) = 50x - 0.5x^2$ and $C(x) = 10x + 200$. What production level x maximizes profit?

a. 40 ✓

b. 50

c. 30

d. 60

Profit $P(x) = R(x) - C(x) = -0.5x^2 + 40x - 200$. $P'(x) = -x + 40 = 0 \Rightarrow x = 40$.

11 Find the point on the line $y = 2x + 1$ that is closest to the point $(4, 4)$.

a. $(2, 5)$ ✓

b. $(1, 3)$

c. $(3, 7)$

d. $(1.5, 4)$

Squared distance
$D(x) = (x-4)^2 + (2x+1-4)^2 = (x-4)^2 + (2x-3)^2 = x^2 - 8x + 16 + 4x^2 - 12x + 9 = 5x^2 - 20x + 25$
. $D'(x) = 10x - 20 = 0 \Rightarrow x = 2, y = 5$.

12 A Norman window has the shape of a rectangle surmounted by a semicircle. If the perimeter is 16 ft, what radius r of the semicircle maximizes the light admitted?

a. $\frac{16}{4+\pi}$ ft ✓

b. $\frac{8}{2+\pi}$ ft

c. 2 ft

d. $\frac{16}{\pi}$ ft

Perimeter: $2r + 2h + \pi r = 16 \Rightarrow 2h = 16 - (2+\pi)r$. Area:
$A = 2rh + \frac{1}{2}\pi r^2 = r(16 - (2+\pi)r) + \frac{1}{2}\pi r^2 = 16r - (2 + \pi/2)r^2$. $A'(r) = 16 - (4+\pi)r = 0 \Rightarrow r = \frac{16}{4+\pi}$ ft.

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